SAT Math · Week 8 — Answer Key and Worked Solutions

Teacher copy · Grade 11 (11C) · do not circulate with the student paper

Conditions. 24 questions, 35 minutes (about 1.45 min each), the student paper carries no preamble. Each question shows its tool: no calculator, calculator, or Desmos. Student-produced responses accept any equivalent exact value; write decimals as printed.

Answer strip. **1**: 42 **2**: (A) **3**: (B) **4**: 2 **5**: (A) **6**: 18 **7**: (B) **8**: (C) **9**: (D) **10**: (D) **11**: (C) **12**: (D) **13**: 120 **14**: (D) **15**: (A) **16**: (A) **17**: (B) **18**: (C) **19**: (C) **20**: 45 **21**: (C) **22**: (A) **23**: 48 **24**: (B)

Balance. Multiple-choice letters (18 items): A 5, B 4, C 5, D 4. Numeric options are printed in ascending order, as on the real test, so their letters follow the value. Student-produced responses: 6 items. Difficulty: Medium 7, Hard 17. Origin: anchor to this week 12, spiral from Weeks 1–7 8, seeded from Week 9 4.

Week 8 anchor (12 items). 8A Angles, Similarity and Right-Triangle Trigonometry, items 9–14 (parallel lines with algebraic angles, a parallel line inside a triangle, ratio of areas of similar triangles, angle of elevation with eye height, area from a trigonometric ratio, exterior angle). 8B Circles: Equations, Arcs, Tangents, Radians, items 15–20 (radius from a general-form equation, tangent and radius, sector area from arc length, equation from a diameter, inscribed and central angles, angular speed in radians). **Spiral (8 items, items 1–8):** an expression question in linear equations (Week 1), a mixture model (Week 2), vertex form from a point (Week 3), vertex time and landing time of a projectile (Week 3), a radical equation with an extraneous root (Week 4), equivalent exponential forms (Week 5), a capacity and rate conversion (Week 5), and the effect of a linear transformation on standard deviation (Week 6). **Seeded (4 items, items 21–24):** sphere in a cylinder, perpendicular lines and intercepts, a square-minus-circle composite shape, and scaling a cylinder (all Week 9). No item or trick repeats Week 7: its spiral and seeds (no-solution equations, two-point line model, strict inequality, absolute value transformation, Vieta, half-life, successive percents, complementary ratios, arc length, cone and cylinder, special right triangles) are not reused.

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1. Answer: 42 HARD · NO CALCULATOR · SPIRAL · WEEK 1

Linear equations in one variable: solve for an expression

THE TRICK — Answer the question asked: the combined expression is already $7x - 7$. Multiply both sides by the common denominator 6: $2(2x - 5) + 3(x + 1) = 42$. Expanding gives $4x - 10 + 3x + 3 = 42$, which simplifies to $7x - 7 = 42$. The value of $7x - 7$ is 42 with no need to find x.

Common error. Students who answer 7 either solve for x ($x = 7$) and stop, or multiply the left side by 6 but not the 7 on the right, which gives $7x - 7 = 7$. Students who answer 6 read the common denominator.

2. Answer: (A) HARD · NO CALCULATOR · SPIRAL · WEEK 2

Linear modeling: mixtures

THE TRICK — Weighted average: the mixture lands closer to the larger share. Let x be the liters of the 20% solution. Then $0.20x + 0.50(30 - x) = 0.40(30)$, so $15 - 0.30x = 12$ and $x = 10$. (Check: 10 liters of 20% and 20 liters of 50% give $2 + 10 = 12$ liters of acid in 30 liters, which is 40%.)

Wrong options. (B) is the liters of acid in the final mixture, $0.40 \cdot 30$. (C) splits the 30 liters equally, which would give 35%. (D) is the liters of the 50% solution, the other unknown, not the one asked for.

3. Answer: (B) HARD · NO CALCULATOR · SPIRAL · WEEK 3

Quadratics: vertex form and a point

THE TRICK — Vertex form: use the extra point to find a. Write $f(x) = a(x - 3)^2 - 4$. The point $(5, 4)$ gives $4 = a(2)^2 - 4$, so $4a = 8$ and $a = 2$. Then $f(0) = 2(0 - 3)^2 - 4 = 18 - 4 = 14$.

Wrong options. (A) takes $a = 1$ without using the point. (C) finds $a = 2$ but forgets to subtract 4. (D) solves $4 = 4a - 4$ as $a = 4$ (dividing 8 by 2 instead of by 4), so $4 \cdot 9 - 4 = 32$.

4. Answer: 2 HARD · NO CALCULATOR · SPIRAL · WEEK 4

Radical equations: extraneous solutions

THE TRICK — Square, solve, then check every root in the original. Square both sides: $3x + 10 = x^2 + 4x + 4$, so $x^2 + x - 6 = 0$ and $(x + 3)(x - 2) = 0$. Check $x = 2$: $\sqrt{16} = 4$ and $2 + 2 = 4$, so it works. Check $x = -3$: $\sqrt{1} = 1$ but $-3 + 2 = -1$, so $x = -3$ is extraneous (a square root is never negative). The only solution is $x = 2$, so the sum is 2.

Common error. Students who answer -1 add both roots of the quadratic, $2 + (-3)$, without checking. Students who answer -3 keep only the extraneous root. Students who answer 1 apply Vieta to $x^2 + x - 6$ and lose the sign.

5. Answer: (A) MEDIUM · NO CALCULATOR · SPIRAL · WEEK 5

Exponential functions: equivalent forms

THE TRICK — $b^{x/n} = \left(b^{1/n}\right)^x$: take the root of the base. $16^{x/4} = \left(16^{1/4}\right)^x = 2^x$, because $2^4 = 16$. So $f(x) = 6 \cdot 2^x$.

Wrong options. (B) divides the base by 4 instead of taking the fourth root, $\frac{16}{4}$ when it should be $\sqrt[4]{16}$. (C) ignores the exponent's denominator. (D) multiplies the base by 4, $16 \cdot 4$.

6. Answer: 18 HARD · NO CALCULATOR · SPIRAL · WEEK 6

One-variable data: effect of a transformation on spread

THE TRICK — Shifting moves the center only; scaling multiplies the spread. Subtracting 5 shifts every value by the same amount, so it does not change the spread. Multiplying by 3 multiplies the standard deviation by 3. The new standard deviation is $3 \cdot 6 = 18$. (The new mean would be $3 \cdot 40 - 5 = 115$.)

Common error. Students who answer 13 apply the whole rule $3 \cdot 6 - 5$ to the standard deviation. Students who answer 6 think scaling does not change spread. Students who answer 115 report the new mean instead.

7. Answer: (B) HARD · CALCULATOR · SPIRAL · WEEK 5 UNITS

Ratios, rates and units: capacity and rate

THE TRICK — Convert the volume first ($1\ \text{m}^3 = 1{,}000$ L), and the minutes last. The volume is $1.2 \cdot 0.5 \cdot 0.9 = 0.54\ \text{m}^3 = 540$ liters. At 4.5 liters per minute the time is $\frac{540}{4.5} = 120$ minutes, which is $\frac{120}{60} = 2$ hours.

Wrong options. (A) uses $1\ \text{m}^3 = 100$ liters, so 54 liters takes 12 minutes, which is 0.2 hour. (C) uses 100 liters and leaves the answer in minutes. (D) is the correct time in minutes, not hours.

8. Answer: (C) HARD · DESMOS · SPIRAL · WEEK 3

Quadratics: vertex time and zeros in context

THE TRICK — Vertex gives the peak time, zero gives the landing time: subtract. The peak is at the vertex, $t = -\frac{48}{2(-16)} = 1.5$ seconds. The ball hits the ground when $h(t) = 0$: $-16(t^2 - 3t - 4) = 0$, so $(t-4)(t+1) = 0$ and $t = 4$ (the root $t = -1$ is before the throw). The time from the peak to the ground is $4 - 1.5 = 2.5$ seconds. (In Desmos, graph h and read the vertex (1.5, 100) and the positive x-intercept (4, 0).)

Wrong options. (A) is the time to reach the peak. (B) halves the total flight time, assuming the path is symmetric about the throw height, which it is not because the ball starts on a platform. (D) is the total time from the throw to the ground.

9. Answer: (D) MEDIUM · NO CALCULATOR · ANCHOR · 8A ANGLES

Lines and angles: parallel lines and a transversal

THE TRICK — Co-interior angles are supplementary, not equal. The two marked angles lie between the parallel lines on the same side of t, so they are co-interior and add to 180°: $(4x + 10) + (2x + 20) = 180$, so $6x = 150$ and $x = 25$. The angles are $4(25) + 10 = 110$ and $2(25) + 20 = 70$. The larger is 110°.

Wrong options. (A) stops at $x = 25$ and never reads the angle. (B) sets the two expressions equal as if they were alternate angles: $4x + 10 = 2x + 20$ gives $x = 5$, and both angles come out as 30°, which cannot be right because the drawing shows one acute and one obtuse angle. (C) is the smaller angle.

10. Answer: (D) MEDIUM · NO CALCULATOR · ANCHOR · 8A SIMILARITY

Similar triangles: parallel line inside a triangle

THE TRICK — Whole over part: the big triangle's side is AB, not DB. Because $\overline{DE} \parallel \overline{BC}$, triangle ADE is similar to triangle ABC. The big triangle's side along $\overline{AB}$ is $AB = 5 + 10 = 15$, so the scale factor is $\frac{15}{5} = 3$. Then $BC = 8 \cdot 3 = 24$.

Wrong options. (A) uses $\frac{AB}{DB} = \frac{15}{10}$, dividing the whole by the wrong part, so $8 \cdot 1.5 = 12$. (B) uses $\frac{DB}{AD} = \frac{10}{5}$, comparing a part with a part, so $8 \cdot 2 = 16$. (C) adds the pieces, $8 + 10$.

11. Answer: (C) HARD · NO CALCULATOR · ANCHOR · 8A SIMILARITY

Similar figures: ratio of areas

THE TRICK — Lengths scale by k, areas scale by k^2. The scale factor from ABC to DEF is $k = \frac{15}{6} = \frac{5}{2}$. Areas scale by $k^2 = \frac{25}{4}$, so the area of DEF is $28 \cdot \frac{25}{4} = 175$.

Wrong options. (A) adds the difference of the sides, $28 + (15 - 6)$. (B) multiplies the area by k itself, $28 \cdot 2.5$, forgetting to square. (D) cubes the scale factor, $28 \cdot 2.5^3$, which is the rule for volumes.

12. Answer: (D) MEDIUM · CALCULATOR · ANCHOR · 8A RIGHT-TRIANGLE TRIG

Right triangles: angle of elevation

THE TRICK — Opposite over adjacent, then add the eye height. The tower rises $40\tan 35°$ above eye level. With the calculator in *degree* mode, $40\tan 35° \approx 28.01$. The eyes are 1.6 m above the ground, so the tower is $28.01 + 1.6 \approx 29.6$ meters tall.

Wrong options. (A) is the result with the calculator in *radian* mode, $40\tan(35) + 1.6$. (B) uses sine instead of tangent: the 40 m is the adjacent side, so sin does not apply. (C) forgets to add the 1.6 m eye height.

13. Answer: 120 HARD · NO CALCULATOR · ANCHOR · 8A RIGHT-TRIANGLE TRIG

Right triangles: trigonometric ratios and area

THE TRICK — Hypotenuse times sine is the opposite leg. $\sin A = \dfrac{BC}{AC}$, so $BC = 26 \cdot \dfrac{5}{13} = 10$. By the Pythagorean theorem, $AB = \sqrt{26^2 - 10^2} = \sqrt{576} = 24$. The legs are perpendicular, so the area is $\dfrac{1}{2} \cdot 10 \cdot 24 = 120$.

Common error. Students who answer 240 multiply the legs and forget the factor $\dfrac{1}{2}$. Students who answer 130 use the hypotenuse as a base, $\dfrac{1}{2} \cdot 26 \cdot 10$. Students who answer 10 or 24 stop at one leg.

14. Answer: (D) MEDIUM · NO CALCULATOR · ANCHOR · 8A ANGLES

Lines and angles: exterior angle of a triangle

THE TRICK — Exterior angle equals the sum of the two remote interior angles. The exterior angle at R equals the two remote interior angles: $7x - 5 = (2x + 10) + (3x + 5)$, so $2x = 20$ and $x = 10$. Then $\angle PRS = 65°$. Angles PRQ and PRS lie on a straight line, so $\angle PRQ = 180 - 65 = 115°$.

Wrong options. (A) stops at $x = 10$. (B) reports $\angle Q = 3(10) + 5$, a different angle. (C) is the exterior angle $\angle PRS$ itself, not the interior angle $\angle PRQ$ the question asks for.

15. Answer: (A) HARD · NO CALCULATOR · ANCHOR · 8B CIRCLES

Circles: equation of a circle by completing the square

THE TRICK — Divide by the leading coefficient first, then complete the square. Divide every term by 2: $x^2 + y^2 - 6x + 8y = 11$. Complete the square: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16$, so $(x-3)^2 + (y+4)^2 = 36$. The radius is $\sqrt{36} = 6$.

Wrong options. (B) completes the square without dividing, adding $9 + 16$ to the undivided 22 (so $r^2 = 47$). (C) is the diameter. (D) is r^2: the number on the right of $(x - h)^2 + (y - k)^2 = r^2$ must be square-rooted.

16. Answer: (A) HARD · NO CALCULATOR · ANCHOR · 8B TANGENTS

Circles: tangent line is perpendicular to the radius

THE TRICK — A tangent meets the radius at 90°: the radius is a height. Because $\overline{PT}$ is tangent at T, $\overline{OT} \perp \overline{PT}$, so triangle OTP has a right angle at T with hypotenuse $OP = 17$. Then $PT = \sqrt{17^2 - 8^2} = \sqrt{225} = 15$. The area is $\dfrac{1}{2} \cdot OT \cdot PT = \dfrac{1}{2} \cdot 8 \cdot 15 = 60$.

Wrong options. (B) uses the hypotenuse as a leg, $\dfrac{1}{2} \cdot 8 \cdot 17$. (C) forgets the factor $\dfrac{1}{2}$, $8 \cdot 15$. (D) combines both slips, $8 \cdot 17$.

17. Answer: (B) HARD · NO CALCULATOR · ANCHOR · 8B ARCS AND SECTORS

Circles: arc length and sector area

THE TRICK — Same fraction of the circle for the arc and the sector. The radius is 12, so the circumference is 24π. The sector is $\dfrac{10\pi}{24\pi} = \dfrac{5}{12}$ of the circle. The circle's area is $\pi(12)^2 = 144\pi$, so the sector's area is $\dfrac{5}{12} \cdot 144\pi = 60\pi$. (Shortcut: area $= \dfrac{1}{2}rs = \dfrac{1}{2} \cdot 12 \cdot 10\pi = 60\pi$.)

Wrong options. (A) is the arc length itself. (C) uses the diameter in the shortcut, $\dfrac{1}{2} \cdot 24 \cdot 10\pi$. (D) uses the diameter as the radius in πr^2 before taking the fraction, $\dfrac{5}{12} \cdot 576\pi$.

18. Answer: (C) MEDIUM · NO CALCULATOR · ANCHOR · 8B CIRCLES

Circles: equation from the endpoints of a diameter

THE TRICK — Midpoint is the center; half the diameter is the radius. The center is the midpoint of the diameter: $\left(\frac{-1+7}{2}, \frac{2+8}{2}\right) = (3, 5)$. The diameter is $\sqrt{(7-(-1))^2 + (8-2)^2} = \sqrt{100} = 10$, so $r = 5$ and $r^2 = 25$. The equation is $(x-3)^2 + (y-5)^2 = 25$.

Wrong options. (A) writes $r = 5$ on the right instead of $r^2 = 25$. (B) has the signs of the center flipped, which places the center at $(-3, -5)$. (D) uses the diameter squared, 10^2, instead of the radius squared.

19. Answer: (C) HARD · NO CALCULATOR · ANCHOR · 8B ARCS AND ANGLES

Circles: inscribed and central angles

THE TRICK — The central angle is double the inscribed angle on the same arc. The central angle is twice the inscribed angle: $7x - 12 = 2(2x + 6) = 4x + 12$, so $3x = 24$ and $x = 8$. Then $\angle ACB = 2(8) + 6 = 22°$ (and $\angle AOB = 7(8) - 12 = 44°$, which is twice $22°$).

Wrong options. (A) stops at $x = 8$. (B) sets the two angles equal, $7x - 12 = 2x + 6$, and gets $x = 3.6$, so $2(3.6) + 6 = 13.2$. (D) is the central angle $\angle AOB$.

20. Answer: 45 HARD · NO CALCULATOR · ANCHOR · 8B RADIANS

Radians: angular speed and rotations

THE TRICK — One full rotation is 2π radians. 2 minutes is 120 seconds, so the wheel turns through $\frac{3\pi}{4} \cdot 120 = 90\pi$ radians. One rotation is 2π radians, so the number of rotations is $\frac{90\pi}{2\pi} = 45$.

Common error. Students who answer 90 divide by π instead of 2π. Students who answer 22.5 divide by 4π. Students who answer 0.75 or 1.5 forget to convert minutes to seconds.

21. Answer: (C) HARD · NO CALCULATOR · SEED · WEEK 9 SOLIDS

Volume: sphere in a cylinder

THE TRICK — Displaced volume equals the sphere's volume. The sphere's volume is $\frac{4}{3}\pi(6)^3 = 288\pi$. The rise h satisfies $\pi(12)^2 h = 288\pi$, so $144h = 288$ and $h = 2$ centimeters.

Wrong options. (A) uses the sphere's surface area $4\pi r^2 = 144\pi$ in place of its volume. (B) drops the factor $\frac{4}{3}$, $\frac{216\pi}{144\pi}$. (D) divides by the base area of a radius-6 circle, $\frac{288\pi}{36\pi}$, instead of the cylinder's.

22. Answer: (A) HARD · NO CALCULATOR · SEED · WEEK 9 COORDINATE GEOMETRY

Coordinate geometry: perpendicular lines and intercepts

THE TRICK — Perpendicular slopes are negative reciprocals. Rewrite $3x + 4y = 12$ as $y = -\frac{3}{4}x + 3$, so its slope is $-\frac{3}{4}$. Line k has slope $\frac{4}{3}$. Using the point $(6, 1)$: $1 = \frac{4}{3}(6) + b$, so $b = 1 - 8 = -7$.

Wrong options. (B) uses the parallel slope $-\frac{3}{4}$: $b = 1 + \frac{3}{4} \cdot 6 = 5.5$. (C) finds $b = -7$ but drops the sign. (D) uses the reciprocal without changing the sign, slope $-\frac{4}{3}$, so $b = 1 + 8 = 9$.

23. Answer: 48 HARD · NO CALCULATOR · SEED · WEEK 9 COMPOSITE SHAPES

Area: composite shapes (square minus inscribed circle)

THE TRICK — Factor the shaded area: square $\times \left(1 - \frac{\pi}{4}\right)$. Let the side of the square be s. The circle has radius $\frac{s}{2}$, so the area outside the circle is $s^2 - \pi\frac{s^2}{4} = \frac{s^2(4-\pi)}{4}$. Setting this equal to $36(4 - \pi)$ gives $\frac{s^2}{4} = 36$, so $s^2 = 144$ and $s = 12$. The perimeter is $4 \cdot 12 = 48$.

Common error. Students who answer 12 stop at the side. Students who answer 144 give the area. Students who answer 24 drop the 4 in the denominator, so $s^2 = 36$ and $s = 6$.

24. Answer: (B) MEDIUM · NO CALCULATOR · SEED · WEEK 9 SOLIDS

Volume: scaling dimensions of a cylinder

THE TRICK — Volume uses the radius squared: the radius factor counts twice. $V = \pi r^2 h$. Tripling r multiplies the volume by $3^2 = 9$; quartering h multiplies it by $\frac{1}{4}$. The new volume is $72 \cdot 9 \cdot \frac{1}{4} = 162$.

Wrong options. (A) multiplies by 3 instead of 9, $72 \cdot 3 \cdot \frac{1}{4}$. (C) is $72 \cdot 3$, with the radius factor unsquared and the height ignored. (D) is $72 \cdot 9$, with the height change ignored.